



Geometric Runge–Gross Theorem

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Geometric TDDFT

Reproducing a given $\rho(t, \mathbf{r})$

$$\begin{cases} i\partial_t \Psi(t) = \left(\hat{T} + \hat{V}(t) + i\hat{W}(t) + \hat{V}_{ee} \right) \Psi(t) \\ \Psi(t=0) = \Psi_0 \end{cases} \implies \rho_{\Psi(t)}(\mathbf{r}) = \rho(t, \mathbf{r})$$

$$\hat{V}(t) = \sum_{j=1}^N V(t, \mathbf{r}_j), \quad \hat{W}(t) = \sum_{j=1}^N W(t, \mathbf{r}_j)$$

- $\langle \Psi(t) | \hat{W}(t) | \Psi(t) \rangle = \int_{\mathbb{R}^d} W(t, \mathbf{r}) \rho(t, \mathbf{r}) d\mathbf{r} = 0$ for all t to ensure norm conservation
- W **not seen in the energy**, really uses the **full complex structure** of TD Schrödinger
- one can reproduce ρ with V or W or both (in practice, V always contains V_{ext})
- W could be useful in Kohn–Sham ($V_{ee} = 0$) to correct adiabatic functionals for V
- W can also be used in Schrödinger ($V_{ee} \neq 0$) to formulate orbital-free TDDFT
- **Runge–Gross theorem** is about uniqueness of V at given W , or conversely

Geometric Runge–Gross theorem

Theorem (Geometric Runge–Gross)

Let $\Psi_1(t)$ and $\Psi_2(t)$ be the solutions to Schrödinger's equation with respective potentials $V_1(t, \mathbf{r}) + iW_1(t, \mathbf{r})$ and $V_2(t, \mathbf{r}) + iW_2(t, \mathbf{r})$. Assume that they have the same initial condition $\Psi_1(0) = \Psi_2(0) = \Psi_0$ and that they yield the same density over some interval of times:

$$\rho_{\Psi_1(t)} = \rho_{\Psi_2(t)} \quad \text{for } t \in [0, T).$$

Under *suitable regularity assumptions* to be specified later, we have the following:

- (a) If $W_1(t, \mathbf{r}) = W_2(t, \mathbf{r})$ for all $t \in [0, T)$, $\mathbf{r} \in \mathbb{R}^d$ and $\ker(K^{\Psi_0}) = \mathbb{C}$, then $V_1(t, \mathbf{r}) = V_2(t, \mathbf{r}) + C(t)$.
- (b) If $V_1(t, \mathbf{r}) = V_2(t, \mathbf{r})$ for all $t \in [0, T)$, $\mathbf{r} \in \mathbb{R}^d$ and $\ker(S^{\Psi_0}) = \{0\}$, then $W_1(t, \mathbf{r}) = W_2(t, \mathbf{r})$.

- operators K^{Ψ_0} and S^{Ψ_0} recalled later
- implicitly assumed that $\langle \Psi_m(t) | \widehat{W}_m(t) | \Psi_m(t) \rangle = 0$
- can also be stated with $i[\widehat{W}_m(t), |\Psi_m(t)\rangle\langle\Psi_m(t)|]$ and then $W_1 = W_2 + C(t)$
- only about **uniqueness**, existence not known in general (see later)

Runge–Gross, *Phys. Rev. Lett.* 1984
Fournais–Lampart–Lewin–Østergaard Sørensen, *Phys. Rev. A* 2016
Cancès–Duez–van Gog–Lauritsen–Lewin–Toulouse, *arXiv* 2026

Application to the universal W^{GKS} functional

► **Universal functional:** $W^{\text{GKS}}[\Psi_0, \Phi_0, \rho] := W_0[\Phi_0, V[\Psi_0, \rho], \rho]$

W reproducing ρ for KS
in an external potential



total V reproducing ρ
in Schrödinger

(Full) Geometric Kohn–Sham equation

$$\begin{cases} i\partial_t \Phi = \left(\hat{T} + \hat{V}_{\text{ext}}(t) + i\widehat{W}^{\text{GKS}}[\Psi_0, \Phi_0, \rho_\Phi] \right) \Phi \\ \Phi(t=0) = \Phi_0 \end{cases}$$

reproduces the exact Schrödinger density in $V_{\text{ext}}(t)$ with initial state Ψ_0

Argument relies on Runge–Gross:

- Uniqueness W in (b) implies $W^{\text{GKS}}[\Psi_0, \Phi_0, \rho_\Phi] = W_0[\Phi_0, V[\Psi_0, \rho_\Phi], \rho_\Phi] = W_0[\Phi_0, V_{\text{ext}}, \rho_\Phi]$
- Uniqueness of V in (a) implies $V[\Psi_0, \rho_\Phi] = V_{\text{ext}} + C(t)$
- Definition of $V[\Psi_0, \rho]$ implies $\rho_\Phi = \rho^S$

Idea of proof of the Runge–Gross theorem

$$\partial_t^{\ell+1} \rho(0) = 2S^{\Psi_0} \partial_t^\ell W(0) + \text{terms involving } \partial_t^j W(0) \text{ and } \partial^\alpha \Psi_0 \text{ for } j \leq \ell \text{ and } |\alpha| \leq 2\ell + 2$$

$$\partial_t^{\ell+2} \rho(0) = K^{\Psi_0} \partial_t^\ell V(0) + \text{terms involving } \partial_t^j V(0) \text{ and } \partial^\alpha \Psi_0 \text{ for } j \leq \ell \text{ and } |\alpha| \leq 2\ell + 4$$

Argue by induction to show that all the time derivatives of $W_1 - W_2$ or $V_1 - V_2$ vanish at $t = 0$

► Easier to argue by **duality, e.g.**

$$\frac{d}{dt} \int_{\mathbb{R}^d} \rho(t, \mathbf{r}) U(\mathbf{r}) d\mathbf{r} = \frac{d}{dt} \langle \Psi(t) | \hat{U} | \Psi(t) \rangle = \langle \Psi(t) | i[\hat{T}, \hat{U}] | \Psi(t) \rangle + 2 \underbrace{\langle \Psi(t) | \hat{U} \hat{W}(t) | \Psi(t) \rangle}_{=:\langle U, S^{\Psi(t)} W(t) \rangle} \quad \text{at } t = 0$$

Subtracting the eqns for Ψ_1 and Ψ_2 and choosing $U = W_1(0) - W_2(0)$ gives $\langle U, S^{\Psi_0} U \rangle = \|\hat{U}\Psi_0\|^2 = 0$

► **Working mathematical assumptions for this proof:**

- V_{ee} infinitely differentiable in space, with bounded derivatives
- Ψ_0 infinitely differentiable in space, with square-integrable derivatives
- V_1, V_2, W_1, W_2 infinitely differentiable in space-time, with bounded space derivatives for all t
- $t \mapsto V_1(t, \mathbf{r}) - V_2(t, \mathbf{r})$ and $t \mapsto W_1(t, \mathbf{r}) - W_2(t, \mathbf{r})$ are real-analytic for all $\mathbf{r}$



Invertibility of K^{Ψ_0} and S^{Ψ_0}

$$\langle V, K^{\Psi} V \rangle = \int_{\mathbb{R}^d} \rho_{\Psi}(\mathbf{r}) |\nabla V(\mathbf{r})|^2 d\mathbf{r}, \quad K^{\Psi} V = -\nabla \cdot (\rho \nabla V)$$

- (unbounded) elliptic operator in divergence form
- $\ker(K^{\Psi}) = \mathbb{C}$ in $L^2(\mathbb{R}^d, \rho)$ if for instance $\rho_{\Psi} > 0$ everywhere (almost everywhere if V continuous)

$$\langle W, S^{\Psi} W \rangle = \|\widehat{W}\Psi\|^2, \quad (S^{\Psi} W)(\mathbf{r}) = \rho_{\Psi}(\mathbf{r}) W(\mathbf{r}) + \int_{\mathbb{R}^d} \rho_{\Psi}^{(2)}(\mathbf{r}, \mathbf{r}') W(\mathbf{r}') d\mathbf{r}'$$

- bounded operator in $L^2(\mathbb{R}^d, \rho)$
- $\ker(S^{\Psi}) = \{0\}$ called **unique representability condition** (Penz-van Leeuwen '25)

$$\sum_{j=1}^N W(\mathbf{r}_j) \Psi = 0 \implies W(\mathbf{r}) \equiv 0$$

valid if γ_{Ψ} acts irreducibly on $\mathbb{R}^d$: $[\gamma_{\Psi}, \mathbb{1}_A] = 0 \implies A = \mathbb{R}^d$

► both properties are satisfied for ground states, by **strong unique continuation** (Garrigue '20)

Difficulties in proving existence of V and W

- ▶ **Runge–Gross proof suggests formulas for $\partial_t^\ell V(0, \mathbf{r})$ or $\partial_t^\ell W(0, \mathbf{r})$ in terms of Ψ_0 only**, hence

$$V(t, \mathbf{r}) = \sum_{\ell \geq 0} \frac{\partial_t^\ell V(0, \mathbf{r})}{\ell!} t^\ell, \quad W(t, \mathbf{r}) = \sum_{\ell \geq 0} \frac{\partial_t^\ell W(0, \mathbf{r})}{\ell!} t^\ell \quad (\text{van Leeuwen '99})$$

- involves high space derivatives of Ψ_0 , not clear if the series is summable (how uniformly in $\mathbf{r}$?)
- expanding Schrödinger's solution like $e^{-itH}\Psi_0 = \sum_{n \geq 0} \frac{(-it)^n H^n}{n!} \Psi_0$ is usually dangerous
- need to invert K^{Ψ_0} or S^{Ψ_0}

- ▶ **Nonlinear Schrödinger equations** (Tokatly '07, Maitra-Todorov-Woodward-Burke '10, Ruggenthaller-Penz-van Leeuwen '11–15)

$$\begin{cases} i\partial_t \Psi(t) = (\hat{T} + \hat{V}(t) + \hat{V}_{ee})\Psi(t) \\ K^{\Psi(t)} \hat{V}(t) = \partial_t^2 \rho + q_4[\Psi(t)] \end{cases}$$

$$\begin{cases} i\partial_t \Psi(t) = (\hat{T} + \hat{V}_{\text{ext}}(t) + i\hat{W}(t) + \hat{V}_{ee})\Psi(t) \\ S^{\Psi(t)} W(t) = (\partial_t \rho + \nabla \cdot \mathbf{j}_\Psi(t))/2 \end{cases}$$

- right approach in finite dimension using Cauchy-Lipschitz theorem ($\approx$ fixed point), cf Éric's talk
- simple fixed point cannot be used easily in continuous case due to **loss of derivatives**
- not covered by standard nonlinear PDE techniques